Thomas Yu

Research Statement

I am an economist working in health economics and industrial organization. My research agenda focuses on the design of public payments in health care markets. Governments spend enormous sums on everything from hospital care to physician training to prescription drugs. These payments often pass through intermediaries that respond strategically, altering the effects of policies at the expense of patients they are meant to help. I apply economic theory and empirical methods, including structural and reduced-form approaches, to evaluate and design policies with the goal of making health care markets more effective, equitable, and efficient.

Current work. In my job market paper, I study public payments directed at hospital obstetric (OB) units, which are closing across the United States and especially in rural areas. Using data on births and medical claims, I estimate a model of the hospital OB care market in Arkansas. I find that the average OB unit generates a positive net externality worth $104 per Arkansas birth, mostly by improving access to care. Mothers highly value having more options and shorter travel distances, but much of this value is not reflected in OB hospitals' profits. A Medicaid price hike keeps OB units open when their fixed costs go up, with a benefit 1.0 to 1.5 times its cost, but it pays more to large hospitals that would have operated OB units anyway. A flat subsidy to all OB hospitals achieves the same goal more cost-effectively, with a benefit 1.2 to 4.2 times its cost.

In my second paper, I study graduate medical education (GME) payments, which incentivize hospitals to train medical residents. About half of teaching hospitals exceed their residency caps and thus receive no additional payments for training more residents. I find that over-cap hospitals do not respond to payments, so redirecting payments from over-cap to under-cap hospitals would lead to more residents without any change in government spending. These gains are proportionally larger in rural areas, which face more severe physician shortages.

My third paper, with classmate Genna Liu, studies drug price regulations that affect how public programs like Medicare pay for prescription drugs. We find that caps on out-of-pocket prices benefit drug users directly, and may reduce total spending, while ceilings on the prices insurers pay do not always reach consumers. We apply our model to Medicare's $35 insulin copay cap and find that the cap increases insulin use without raising costs for other consumers.

A common theme in these papers is that public payments change behavior in equilibrium, and understanding when behavior changes, and when it does not, leads to better policy design. My works in progress extend my research agenda in three broad areas:

Access to care. Health care can only provide value if it is accessible: a hospital that delivers babies, an emergency room that is open, or a specialist who can be reached. Public payments allow care to exist in places where it otherwise would not, but they are not always well-targeted. I want to understand how public payments should promote access, including which services are worth sustaining, in which places, and through which policy instruments. My job market paper on OB units fits into this area. An ongoing project extends it to study rural hospitals' full sets of service lines; a hospital decides which services to keep as a bundle, and the payments for one affect whether it keeps the rest. Another project studies the adoption of technologies such as telehealth and artificial intelligence, which can expand the reach of high-quality care. I examine whether

incentives for adoption simply reward providers that already see many patients, or whether they spread technologies to where they are needed.

The health care workforce. Who delivers care is itself a product of public payments. Governments subsidize training, pay different rates for different types of providers, and decide what each type may do. Provider shortages are concentrated in primary care and in rural areas, but existing payments are often blind to specialty, place, and provider type. I consider how policies can produce the specific providers that a health care system needs. One project builds on my paper on GME payments by studying how these payments affect which specialties hospitals choose to train and where their residents end up practicing. Another project studies non-physician providers, like physician assistants and nurse practitioners, who are paid less than physicians and restricted through scope-of-practice and immigration laws. I evaluate whether raising payments or easing restrictions for these providers can improve access to care, and whether there are long-run effects on prices, insurance networks, and health outcomes.

The incidence of health policy. Many public payments do not go directly to providers, but rather upstream firms such as insurers and drug manufacturers. Their equilibrium effects can be hard to predict, and the share of benefits that reaches patients is decided by market structure and complex firm relationships. I want to map how policies affect firms and patients, so they can be best designed to reach the people they target. In a project with Genna Liu, related to our paper on drug price regulations, we study the public negotiation of drug prices and how negotiation terms can be structured to prioritize lower premiums or out-of-pocket costs for consumers. Another project studies insurance premium subsidies, which are meant to support enrollees but may have indirect effects on the prices insurers pay providers through bargaining.

Health care systems are sustained by vast public payments to hospitals, physicians, insurers, and drug manufacturers, each of whom has their own incentives. Too often, there is a gap between what governments pay for and what patients receive. The goal of my research is to close that gap, ensuring that public money buys the care that people need most.